Fixed-core response to a radial quartic phase in a lens-free scalar-paraxial window: small approximately family-independent changes and an unresolved caustic-specific residual

Auto BeamScientist Project

Abstract

We report an evidence-bound numerical study of how a rotationally symmetric quartic phase changes fixed-core efficiency in a lens-free scalar-paraxial window. The model-authored question asks whether, for centrosymmetric inputs, the change is fully accounted for by the intensity-moment equivalent quadratic defocus $q_{2,\mathrm{eff}} = a_4\langle(r/w)^4\rangle/\langle(r/w)^2\rangle$, while non-centrosymmetric caustic fields retain a residual beyond that zero-parameter prediction. We executed five primary free-space cases with a conservative angular-spectrum propagator, no lens, zero rotation, zero relative phase, and unit amplitude weight: Gaussian, Pearcey, Butterfly, Airy, and a matched-curvature chirp-plus-ripple Gaussian control. At quartic strength $a_4 = 0.7$, the fixed-core efficiency decreased relative to the corresponding base arm by 1.60% (Gaussian), 2.21% (Pearcey), and 1.66% (Butterfly). The Airy arm changed by 2.81% but failed its sampling-retention and window-edge diagnostics and is therefore not testable as configured. The matched-curvature control changed the same observable by 29.0%, roughly an order of magnitude more than the quartic term. The pre-registered moment-equivalent quadratic arm was not executed, so the discriminating residual R is undefined; the recorded 5% convergence tolerance is also coarser than the measured quartic effect. The data therefore neither support a caustic-specific residual nor establish second-moment universality. They bound a small, same-sign, approximately family-independent quartic response within the executed window and leave the central mechanism unresolved.

1 Introduction

Structured phase elements provide a practical route to reshape the transverse structure and axial evolution of coherent light. Canonical caustic fields such as Pearcey, Butterfly, and finite-energy Airy beams have attracted attention because their transverse phase structure produces curved high-intensity features, auto-focusing, and self-healing behavior [5, 11, 1, 2]. In parallel, phase-only reshaping is used to modify beam quality and spatial-mode content [7, 3]. The present study concerns a different and deliberately simpler object: a rotationally symmetric quartic phase mask applied to a scalar input field in free space. The mask is not a canonical catastrophe beam; it is an aberration-like phase transformation whose axial effect is to be compared with a moment-equivalent quadratic defocus.

The model-selected question is:

> *In a lens-free scalar-paraxial window, is the change in fixed-core on-axis concentration caused by a rotationally symmetric quartic phase fully accounted for by its intensity-moment equivalent quadratic defocus $q_{2,\mathrm{eff}} = a_4\langle(r/w)^4\rangle/\langle(r/w)^2\rangle$ for centrosymmetric inputs, while non-centrosymmetric caustics (Pearcey, Butterfly, Airy) show a residual beyond that zero-parameter prediction?*

The registered observable is `core_efficiency`, the circular-core energy fraction defined below and evaluated on a fixed axial window. The comparison is between a transformed arm and its base arm under fixed power, aperture, grid, and propagation geometry.

This manuscript reports what was actually executed. Five primary arms were run: Gaussian, Pearcey, Butterfly, Airy, and a matched-curvature non-centrosymmetric Gaussian control. The quartic strength was $a_4 = 0.7$. A conservative angular-spectrum propagator was used with no lens, free-space environment, phase-noise root-mean-square 0.02, and composition parameters $\theta = 0$, $\delta = 0$, $a = 1$. Confirmation records varied mesh size, window size, propagator, wavelength, axial sampling, and a quantization diagnostic. The central finding is deliberately modest: in the three arms that passed their primary diagnostics, the quartic phase suppressed fixed-core efficiency by 1.6–2.2%, with the same sign and comparable magnitude across Gaussian, Pearcey, and Butterfly inputs. The moment-equivalent null arm required by the discriminating statistic was not executed, and the Airy arm failed its own sampling/window gates. The matched-curvature control produced a much larger 29.0% change, showing that generic phase shaping can dominate the quartic effect in this regime.

The paper is organized as follows. Section 2 places the work in relation to caustic-beam and propagation literature. Section 3 defines the input normalization, propagation, composition, observable, and moment-equivalent residual. Section 4 gives the executed numerical protocol and parameter values. Section 5 reports the primary and confirmation results, including the missing null arm and the invalid Airy arm. Section 6 discusses physical interpretation and limits. Section 7 concludes.

2 Related Work

Pearcey beams are canonical cusp-caustic fields with auto-focusing and self-healing properties [5]. Symmetric and masked Pearcey-Gaussian constructions extend this family and control the axial caustic structure [9, 8]. Butterfly beams are higher-order optical catastrophe fields whose caustic geometry is richer than a simple quadratic focus [11]. Finite-energy Airy beams provide an accelerating, non-canonical caustic-like alternative [1, 2]. Propagation of Pearcey fields has also been analyzed in anisotropic media [10]. These studies establish the beam families and the qualitative axial structures, but they do not supply the moment-equivalent null comparison used here.

Numerically, free-space scalar propagation is commonly performed with angular-spectrum or Fresnel methods. The angular-spectrum method requires attention to evanescent and high-spatial-frequency content; the Matsushima–Shimobaba band-limited construction is a standard reference [4]. In this work we use a conservative radial cutoff in the angular-spectrum transfer function. This cutoff is a numerical safeguard, not the complete Matsushima construction.

Phase-only reshaping and beam-quality control are well-developed topics [7]. The quartic phase considered here is a low-order rotationally symmetric phase mask. Standard aberration theory already suggests that, near a focus, a quartic phase can act partly like an effective defocus. The present study does not claim novelty for that general idea. Its question is narrower: whether a zero-free-parameter moment-equivalent quadratic arm can account for the fixed-core response, and whether caustic inputs show a measurable residual. The executed evidence does not complete that test.

3 Theoretical model

3.1 Input field and normalization

The scalar input is sampled on a Cartesian grid and normalized to power P:

$$U_0(x_j, y_k) = \sqrt{\frac{P}{\Delta x^2 \sum_{a,b} |V_{ab}|^2}}\, V(x_j, y_k), \tag{1}$$

where V_{ab} are the sampled input amplitudes and Δx is the grid spacing. This normalization fixes the total power and makes core-efficiency values comparable across phase-only transformations.

For the Airy arm,

$$V(x,y) = \mathrm{Ai}\Big(\frac{x}{s}\Big)\,\mathrm{Ai}\Big(\frac{y}{s}\Big)\exp\left[\alpha\frac{x+y}{s}\right], \tag{2}$$

inside the aperture, with scale s and apodization parameter α. In the executed Airy case, $s = 1.0 \times 10^{-4}\,\mathrm{m}$ and $\alpha = 0.1$.

For the Pearcey and Butterfly arms,

$$V(x,y) = C_n\Big(\frac{x}{s},\frac{y}{s}\Big)\exp\left(-\frac{r^2}{w^2}\right), \qquad C_n(u,v) = \int \exp\big[i\left(t^n + vt^2 + ut\right)\big]\,dt, \tag{3}$$

with $n = 4$ for Pearcey and $n = 6$ for Butterfly. The Gaussian envelope radius w comes from the optical setup; here $w = 3.0 \times 10^{-4}\,\mathrm{m}$. For the Butterfly arm, the unused Butterfly control coefficients were set to zero. The Gaussian arms use the same setup waist. The recorded apodization field affects the Airy construction through α; for the Pearcey and Butterfly entries it is retained as metadata but does not modify the Gaussian envelope in Eq. (3).

A rotationally symmetric quartic phase is applied as

$$U_q(x,y) = U_0(x,y)\exp\left[ia_4\left(\frac{r}{w}\right)^4\right], \qquad r^2 = x^2 + y^2, \tag{4}$$

with $a_4 = 0.7$ in the primary quartic arms. This is a phase-only mask; it does not change the normalized power. The matched-curvature control uses a quadratic chirp of numerical amount $a_2 = 0.7$ plus a bounded deterministic ripple of amplitude 0.5. The quartic mask is not a canonical catastrophe beam.

3.2 Propagation and composition

Propagation over distance z is computed with the angular-spectrum method:

$$U_z = \mathcal{F}^{-1}\Big[\mathcal{F}(U_0)\,B_z\,\exp\Big(i2\pi z\sqrt{\lambda^{-2} - f_x^2 - f_y^2}\Big)\Big], \tag{5}$$

where λ is the wavelength and B_z is a conservative radial band-limit factor. This factor suppresses evanescent and high-spatial-frequency components. It is not the complete Matsushima band-limited construction.

The composition of two coherent arms is written with an explicit rotation θ, relative phase δ, and amplitude weight a:

$$U_c = \sqrt{\frac{P}{\int |U + ae^{i\delta}\mathcal{R}_\theta U|^2\,dA}}\,\Big(U + ae^{i\delta}\mathcal{R}_\theta U\Big). \tag{6}$$

Here $\mathcal{R}_\theta$ denotes rotation by θ. In every executed primary case the composition was $\theta = 0$, $\delta = 0$, and $a = 1$. Consequently the rotated and coherent arms carry the same transverse intensity as the single arm under the same normalization. An exactly zero coherent field cannot be normalized by Eq. (6); in that case the coherent-arm value is undefined, while the other arms remain measurable. No coherent cancellation occurred in the executed cases.

3.3 Observable and moment-equivalent prediction

The registered observable is the circular-core energy fraction

$$C(z) = P^{-1}\int_{x^2+y^2\le r_c^2} |U_z|^2\,dA, \qquad \overline{C} = \mathrm{median}_{z\in\mathcal{Z}}\,C(z), \tag{7}$$

ID	family	s (m)	apodization	transformation	seed
C0001H001	Gaussian	2.0×10^{-4}	0.3	quartic $a_4 = 0.7$	101
C0001H002	Pearcey	2.0×10^{-4}	0.3	quartic $a_4 = 0.7$	202
C0001H003	Butterfly	2.0×10^{-4}	0.3	quartic $a_4 = 0.7$	303
C0001H004	Airy	1.0×10^{-4}	0.1	quartic $a_4 = 0.7$	404
C0001H005	Gaussian	2.0×10^{-4}	0.3	chirp $a_2 = 0.7$ + ripple 0.5	505

Table 1: Primary hypothesis arms actually executed. All used $n = 192$, extent 2.4×10^{-3} m, $\lambda = 6.33 \times 10^{-7}$ m, waist 3.0×10^{-4} m, power 1.0 W, no lens, angular-spectrum propagation, free-space phase noise 0.02, and composition $\theta = 0$, $\delta = 0$, $a = 1$.

where $r_c = 5.0 \times 10^{-5}$ m and $\mathcal{Z}$ is the fixed axial window from $z = 0.02$ m to $z = 0.08$ m with 13 planes. We denote the resulting core efficiency by $\eta = \overline{C}$.

For a frozen sampled input intensity $I(x, y) = |U_0(x, y)|^2$, the moment ratio is

$$M = \frac{\sum_{j,k} I_{jk} \, (r_{jk}/w)^4}{\sum_{j,k} I_{jk} \, (r_{jk}/w)^2}, \qquad q_{2,\mathrm{eff}} = a_4 M. \tag{8}$$

The zero-free-parameter equivalent-quadratic arm is defined by using $a_2 = q_{2,\mathrm{eff}}$ in place of the quartic phase. The pre-registered normalized residual is

$$R = \frac{|\eta_{\mathrm{quartic}} - \eta_{\mathrm{equiv}}|}{|\eta_{\mathrm{equiv}} - \eta_{\mathrm{base}}|}, \tag{9}$$

provided the denominator is nonzero. The protocol's bands were $R \leq 0.15$ for the Gaussian anchor, $R > 0.25$ for a supported caustic-specific residual in Pearcey and Butterfly, and $R \leq 0.10$ for a retained bounded negative.

No equivalent-quadratic arm was executed in any case. Therefore R was not computed, and the reported comparison values are quartic-versus-base only. The values of M, $q_{2,\mathrm{eff}}$, their discretization errors, and the residual quadratic content of the quartic arm were not logged. This is the central evidential limitation of the study.

4 Numerical methods and executed protocol

The primary optical setup was an $n = 192$ grid over an extent of 2.4×10^{-3} m, wavelength $\lambda = 6.33 \times 10^{-7}$ m, waist $w = 3.0 \times 10^{-4}$ m, listed focal length 0.06 m, and power $P = 1.0$ W. The lens was not applied: `lens_applied=false`. The propagator was the angular-spectrum method; the environment was free space with phase-noise root-mean-square 0.02. The axial planes were $z = 0.020, 0.025, \ldots, 0.080$ m (13 planes). The composition was $\theta = 0$, $\delta = 0$, $a = 1$ in every case.

Table 1 lists the five primary hypothesis arms. The Gaussian, Pearcey, and Butterfly arms used scale $s = 2.0 \times 10^{-4}$ m and recorded apodization 0.3. The Airy arm used $s = 1.0 \times 10^{-4}$ m and apodization 0.1. The quartic strength was $a_4 = 0.7$. The matched-curvature control used a Gaussian with chirp $a_2 = 0.7$ and ripple 0.5.

Three classical controls were optimized over 32 attempts each with identical power, aperture, grid, and no lens: a defocus family with $q_2 = 1.2$, a quadratic-plus-quartic family with $q_2 = 1.2$ and $q_4 = 0.7$, and a four-parameter family that collapsed to the same point. Their reported values are absolute registered observables, not negated optimizer scores.

Diagnostic checks included finite fields, equal power, sampling retention, and window-edge fraction. An arm that failed a check was retained as an executed diagnostic record but was not treated as positive evidence. Confirmation conditions were mesh256, mesh384, Fresnel384, window576, wavelength640, axial41, and quantization. The quantization condition failed sampling retention and is not used as evidence. The Airy arm failed sampling retention and window-edge checks in the primary run and remains untestable as configured.

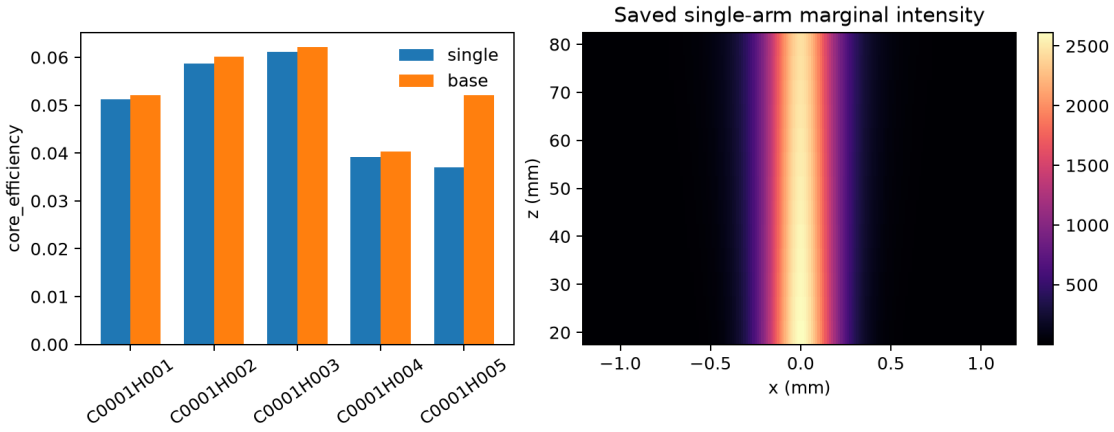


Figure 1: Overview of the executed numerical cases. The figure summarizes the primary arms and the matched-curvature control; all numerical claims in the text are taken from the recorded tables. The Airy arm is marked as untestable because it failed sampling-retention and window-edge diagnostics.

ID	family	base η	transformed η	relative change	primary diagnostics
C0001H001	Gaussian	0.0520747	0.0512417	-1.60%	pass
C0001H002	Pearcey	0.0600749	0.0587485	-2.21%	pass
C0001H003	Butterfly	0.0621276	0.0610955	-1.66%	pass
C0001H004	Airy	0.0403029	0.0391722	-2.81%	sampling/window fail
C0001H005	Gaussian	0.0520747	0.0369704	-29.00%	pass

Table 2: Primary core-efficiency values from the executed records. The transformed arm is quartic $a_4 = 0.7$ for H001–H004 and chirp $a_2 = 0.7$ plus ripple 0.5 for H005. Relative change is $(\eta_{\text{transformed}} - \eta_{\text{base}})/\eta_{\text{base}}$. H004 failed sampling-retention and window-edge diagnostics and is not used as evidence for a caustic residual.

5 Results and discussion

Figure 1 provides an overview of the executed case set. The quantitative statements below are taken from the recorded tables, not estimated from the rendering.

5.1 Primary quartic response

Table 2 lists the base and transformed core efficiencies and the relative change $(\eta_{\text{transformed}} - \eta_{\text{base}})/\eta_{\text{base}}$. The absolute base values differ among families because the input transverse shapes differ; we do not interpret those differences as absolute concentration or compare them as a cross-family concentration ranking.

For the three arms that passed their primary diagnostics, the quartic phase reduced fixed-core efficiency by 1.60% (Gaussian), 2.21% (Pearcey), and 1.66% (Butterfly). The sign is the same in all three cases. The spread in magnitudes is about 0.61 percentage points. No arm shows a 25% or larger caustic-sized residual. The Gaussian and Butterfly changes are nearly equal, while the Pearcey change is only about 0.6 percentage points larger in magnitude. This is not the pattern predicted by a strongly caustic-specific residual of the pre-registered size.

The matched-curvature Gaussian control (H005) changed the same observable by -29.00%. Its primary diagnostics passed. This control uses a quadratic chirp of numerical amount 0.7 plus a bounded deterministic ripple. It shows that a non-centrosymmetric phase-shaped Gaussian can produce a much larger fixed-core change than the quartic mask. The control therefore does not isolate a caustic-specific mechanism; it demonstrates that generic phase shaping can dominate

the quartic response in this window.

5.2 Confirmation and numerical stability

The Gaussian quartic effect was stable under the listed confirmation conditions. At mesh256 the relative change was -1.61%; at mesh384, Fresnel384, window576, and axial41 it was -1.60%; at wavelength640 it was -1.69%. The quantization diagnostic failed sampling retention and gave $+8.41\%$, so it is not used as positive evidence.

The Pearcey quartic effect was likewise stable: -2.22% at mesh256, -2.21% at mesh384, Fresnel384, window576, and axial41, and -2.29% at wavelength640. The Pearcey quantization diagnostic failed sampling retention and integral convergence and gave -10.76%; it is not used as evidence. No confirmation series is available for the Butterfly arm in the supplied records.

The Airy arm changed by -2.81% in the primary comparison. However, its minimum output power was 0.9424 against a retained-power requirement of 0.995, and its maximum edge fraction was 0.1105 against a window budget of 10^{-3} (base arm 0.0317). The arm failed both sampling retention and window-edge checks. It is therefore untestable as configured. Its numerical value is reported as a diagnostic, not as evidence for or against a caustic-specific residual. Changing the window or apodization after seeing this failure would be a post-hoc design change and is not done here.

The confirmation records show that the small Gaussian and Pearcey shifts are reproducible under the listed mesh, window, propagator, wavelength, and axial-sampling changes. They do not, however, make the pre-registered discrimination test executable. The protocol's recorded 5% grid-convergence tolerance is coarser than the measured 1.6–2.2% effect. Under that protocol, the quartic signal is below the stated resolution budget even though the raw comparison is stable. A small stable shift is observable in the raw records, but it cannot be used as a resolved discriminator at the pre-registered tolerance.

5.3 Controls and the missing null arm

The optimized defocus control with $q_2 = 1.2$ reached an absolute core efficiency of 0.0408236. Adding quartic freedom at $q_2 = 1.2$, $q_4 = 0.7$ changed that value to 0.0403671, a relative increment of -1.12%. The four-parameter family collapsed to the same point. This is consistent with the quartic term acting like a small extra defocus on a Gaussian, but it tests no caustic family and cannot substitute for the absent moment-equivalent arm.

The decisive null arm was never executed. No case contains an equivalent-quadratic arm with $a_2 = a_4 M$ computed from the frozen sampled input. The reported comparison values in the hypothesis records are quartic-versus-base, not quartic-versus-equivalent. Therefore the pre-registered residual R in Eq. (9) is undefined, and the prediction-supported flags in the records are not evidence about the mechanism. The missing arm also means that the moment ratio M, its discretization error, and the residual quadratic content were not logged. The study therefore cannot decide whether the small quartic response is fully accounted for by the moment-equivalent defocus.

In every primary case, the rotated, coherent, and single arms are numerically identical because $\theta = 0$, $\delta = 0$, and $a = 1$. The intensity overlap of the rotated and single arms is 1.000000 in the records. Thus the rotated branch provides no independent symmetry check for these centrosymmetric inputs. This redundancy is not a positive control.

6 Physical interpretation and limits

A radial quartic phase adds a transverse phase gradient that is not linear in r. Its leading axial effect on a symmetric field can be represented, to low order, as an equivalent quadratic defocus whose strength is fixed by the input's radial intensity moments. For a Gaussian, the

complex q-parameter is exhausted by second moments, so a quartic phase and its moment-equivalent quadratic arm would be expected to agree in core efficiency within numerical error. That expectation is the Gaussian anchor of the hypothesis. It was not tested here because the equivalent-quadratic arm was not run.

For canonical cusp or accelerating caustic fields, the axial structure is governed by stationary-phase ray convergence and higher-order phase structure. A second-moment reduction would not be expected to encode all of that structure. That expectation is the caustic-specific half of the hypothesis. The present data do not show a caustic-specific residual of the predicted size. At $a_4 = 0.7$, Gaussian, Pearcey, and Butterfly all show a small same-sign suppression of fixed-core efficiency, with magnitudes between 1.6% and 2.2%. The Airy arm is invalid as configured. The matched-curvature control shows that a different phase-shaped Gaussian can change the same observable by 29.0%, which is far larger than the quartic effect.

The most defensible interpretation is therefore a bounded small-effect statement. Within this lens-free scalar-paraxial window, at $a_4 = 0.7$ and $n = 192$, the quartic phase suppresses fixed-core efficiency by roughly 1.6–2.2%, with the same sign for Gaussian, Pearcey, and Butterfly inputs. No caustic-specific anomaly of the pre-registered 25% size is resolvable. The data do not establish second-moment universality, because the equivalent-quadratic arm is missing. They also do not refute it, because the effect is below the protocol's own convergence tolerance and the null statistic is undefined.

Several limits follow directly. First, the quartic term is a weak perturbation in this parameter regime. The primary effect is smaller than the protocol's accepted 5% grid-convergence budget, so even a correctly built equivalent arm would face an instrument-limited comparison at this tolerance. Second, the matched-curvature control is much larger than the quartic effect; generic phase shaping can dominate, and the quartic response is not isolated by the present control. Third, the Airy arm fails its own admission gates, so the frozen 2.4 mm window is inadequate for that family. Fourth, the rotated branch is degenerate at $\theta = 0$, so no independent null check was performed. Fifth, the optimizer controls report absolute observables in separate runs; they are not negated scores and do not test caustic geometry.

The physical picture consistent with the executed data is that the quartic mask acts as a weak aberration-like phase in this window, producing a small reduction of core energy without a visible family-specific axial signature at the 1.6–2.2% level. That picture is a description of the measured window, not a confirmed mechanism. A stronger statement would require the missing equivalent-quadratic arm, a logged moment audit, and a numerical tolerance at least an order of magnitude below the expected effect.

7 Conclusions

The model-selected question remains unresolved by the executed evidence. At quartic strength $a_4 = 0.7$, in a lens-free scalar-paraxial window with no applied lens, free-space propagation, composition $\theta = 0$, $\delta = 0$, $a = 1$, and a conservative angular-spectrum propagator, the fixed-core efficiency decreased by 1.60% for a Gaussian input, 2.21% for Pearcey, and 1.66% for Butterfly. The Airy arm changed by 2.81% but failed sampling-retention and window-edge diagnostics and is untestable as configured. The matched-curvature chirp-plus-ripple control changed the same observable by 29.0%, showing that generic phase shaping can be much larger than the quartic term.

The pre-registered moment-equivalent quadratic arm was not executed in any case. Consequently the normalized residual R was not computed, and no caustic-specific residual can be claimed or refuted. The raw quartic effects are the same sign and comparable magnitude across Gaussian, Pearcey, and Butterfly, which does not resemble the predicted 25% caustic-specific anomaly. However, the effect is below the protocol's own 5% convergence tolerance, so the present instrument cannot adjudicate the mechanism at the pre-registered precision.

The defensible conclusion is a bounded negative: within this window, the quartic phase produces a small, approximately family-independent suppression of fixed-core efficiency, with no resolvable caustic-specific residual. The data give no support for a strong second-moment universality claim either, because the null arm is absent. A future discriminating test would need the zero-free-parameter equivalent-quadratic arm, a logged moment audit, and a pre-registered precision at least ten times below the observed effect. The Airy family would require a fresh window and edge budget rather than reuse of the frozen 2.4 mm window. No novelty claim about quartic-as-effective-defocus is made here.

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Disclosures

The authors declare no financial conflicts of interest. This study reports numerical propagation; no physical experiment is reported.

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